

My sql triggers 2520030372

AIM

To study and implement MySQL Triggers using MySQL Workbench and understand how triggers automatically execute when **INSERT**, **UPDATE**, or **DELETE** operations are performed on a table.

OBJECTIVES

1. To understand the concept of MySQL Triggers.
2. To implement **AFTER INSERT**, **AFTER UPDATE**, and **AFTER DELETE** triggers.
3. To implement **BEFORE INSERT** and **BEFORE UPDATE** triggers.
4. To understand the use of **NEW** and **OLD** values in triggers.
5. To use triggers for auditing, validation, and enforcing business rules.

SOFTWARE REQUIREMENTS

- MySQL Workbench
- MySQL Server

THEORY

A Trigger is a database object that automatically executes when a specified event occurs on a table. A trigger is not called manually; MySQL automatically executes it when an **INSERT**, **UPDATE**, or **DELETE** operation occurs.

Triggers are mainly used for:

- Auditing
- Validation
- Automatic calculations
- Maintaining related or summary data
- Logging **INSERT**, **UPDATE**, and **DELETE** operations
- Enforcing business rules

TYPES OF MYSQL TRIGGERS

Timing

1. BEFORE

2. AFTER

Events

1. INSERT
2. UPDATE
3. DELETE

TABLES USED

Employees Table

The `employees` table stores employee details such as employee ID, employee name, department ID, and salary.

Employee Audit Table

The `employee_audit` table stores information about actions performed on employee records, such as INSERT, UPDATE, and DELETE operations.

TRIGGERS IMPLEMENTED

1. AFTER INSERT Trigger

The `after_employee_insert` trigger automatically records a new employee in the `employee_audit` table whenever an employee is inserted into the `employees` table.

2. AFTER DELETE Trigger

The `after_employee_delete` trigger automatically records the details of an employee when the employee is deleted from the `employees` table.

3. AFTER UPDATE Trigger

The `after_employee_update` trigger records an update whenever employee information is modified.

4. BEFORE INSERT Trigger

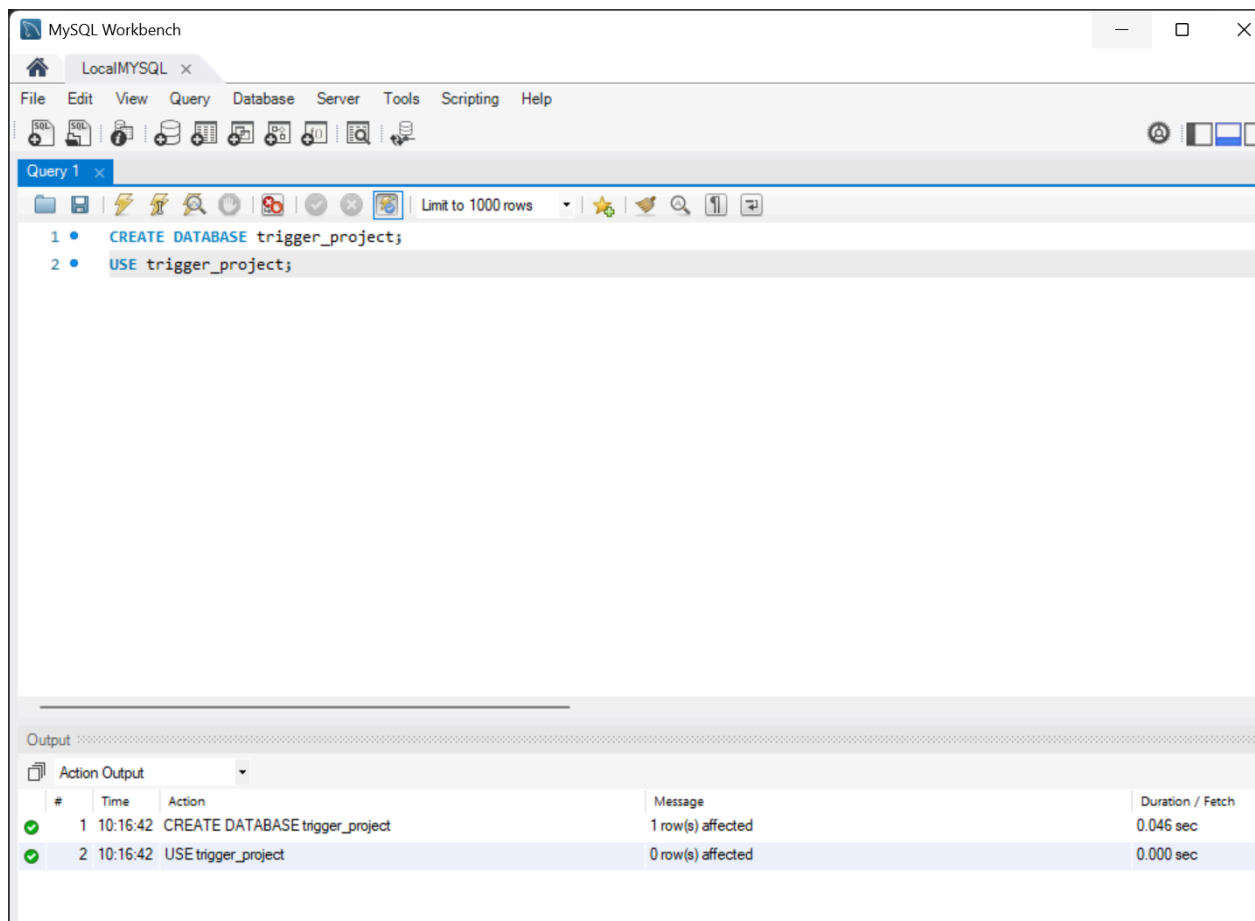
The `check_salary` trigger validates the employee salary before insertion. It prevents an employee from being inserted if the salary is less than ₹20,000.

5. BEFORE UPDATE Trigger

The `prevent_salary_reduction` trigger prevents the salary of an employee from being reduced during an update.

CONCLUSION / RESULT

Thus, MySQL Triggers were successfully studied and implemented using MySQL Workbench. The triggers were used for automatic auditing, validation, logging operations, and enforcing business rules on the employee database.



MySQL Workbench

LocalMySQL x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
1 • CREATE DATABASE trigger_project;
2 • USE trigger_project;
3 • CREATE TABLE employees (
4     emp_id INT PRIMARY KEY,
5     emp_name VARCHAR(50),
6     dept_id INT,
7     salary DECIMAL(10,2)
8 );
9
10 • CREATE TABLE employee_audit (
11     audit_id INT AUTO_INCREMENT PRIMARY KEY,
12     emp_id INT,
13     emp_name VARCHAR(50),
14     action_type VARCHAR(20),
15     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
16 );
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 1	10:16:42	CREATE DATABASE trigger_project	1 row(s) affected	0.046 sec
✓ 2	10:16:42	USE trigger_project	0 row(s) affected	0.000 sec
✗ 3	10:17:42	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
✓ 4	10:17:42	USE trigger_project	0 row(s) affected	0.000 sec
✓ 5	10:17:42	CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_nam...	0 row(s) affected	0.062 sec
✓ 6	10:17:42	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT P...	0 row(s) affected	0.032 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

Limit to 1000 rows

```
6      dept_id INT,
7      salary DECIMAL(10,2)
8  );
9
10 • CREATE TABLE employee_audit (
11     audit_id INT AUTO_INCREMENT PRIMARY KEY,
12     emp_id INT,
13     emp_name VARCHAR(50),
14     action_type VARCHAR(20),
15     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
16 );
17 • INSERT INTO employees VALUES
18 (101, 'Rahul', 1, 50000),
19 (102, 'Priya', 2, 45000),
20 (103, 'Kiran', 1, 55000),
21 (104, 'Anjali', 3, 48000),
22 (105, 'Vijay', 2, 52000);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✗	1 14:42:55	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
✗	2 14:43:02	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
✓	3 14:43:02	USE trigger_project	0 row(s) affected	0.000 sec
✗	4 14:43:02	CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_nam...	Error Code: 1050. Table 'employees' already exists	0.031 sec
✗	5 14:43:03	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT P...	Error Code: 1050. Table 'employee_audit' already exists	0.000 sec
✓	6 14:43:03	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya', 2...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

MySQL Workbench

LocalMySQL x

File Edit View Query Database Server Tools Scripting Help

Limit to 1000 rows

```
7 salary DECIMAL(10,2);
8 };
9
10 CREATE TABLE employee_audit (
11     audit_id INT AUTO_INCREMENT PRIMARY KEY,
12     emp_id INT,
13     emp_name VARCHAR(50),
14     action_type VARCHAR(20),
15     action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
16 );
17 INSERT INTO employees VALUES
18 (101, 'Rahul', 1, 50000),
19 (102, 'Priya', 2, 45000),
20 (103, 'Kiran', 1, 55000),
21 (104, 'Anjali', 3, 48000),
22 (105, 'Vijay', 2, 52000);
23 SELECT * FROM employees;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	14:43:02	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
3	14:43:02	USE trigger_project	0 row(s) affected	0.000 sec
4	14:43:02	CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_na...	Error Code: 1050. Table 'employees' already exists	0.031 sec
5	14:43:03	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT ...	Error Code: 1050. Table 'employee_audit' already exists	0.000 sec
6	14:43:03	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya'...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
7	14:43:43	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers* x

Limit to 1000 rows

```

17 • INSERT INTO employees VALUES
18 (101, 'Rahul', 1, 50000),
19 (102, 'Priya', 2, 45000),
20 (103, 'Kiran', 1, 55000),
21 (104, 'Anjali', 3, 48000),
22 (105, 'Vijay', 2, 52000);
23 • SELECT * FROM employees;

```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	45000.00
103	Kiran	1	55000.00
104	Anjali	3	48000.00
105	Vijay	2	52000.00
NULL	NULL	NULL	NULL

employees 1 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	14:44:22	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
9	14:44:22	USE trigger_project	0 row(s) affected	0.000 sec
10	14:44:22	CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_n...	Error Code: 1050. Table 'employees' already exists	0.000 sec
11	14:44:22	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT...	Error Code: 1050. Table 'employee_audit' already exists	0.000 sec
12	14:44:22	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec
13	14:44:22	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```

30 INSERT INTO employee_audit
31 (emp_id, emp_name, action_type)
32 VALUES
33 (NEW.emp_id, NEW.emp_name, 'INSERT');
34 END $$
35
36 DELIMITER ;

```

Result Grid

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	45000.00
103	Kiran	1	55000.00
104	Anjali	3	48000.00
105	Vijay	2	52000.00
NULL	NULL	NULL	NULL

employees 3 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
23	14:46:50	USE trigger_project	0 row(s) affected	0.000 sec
24	14:46:50	CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_n...	Error Code: 1050. Table 'employees' already exists	0.016 sec
25	14:46:50	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT...	Error Code: 1050. Table 'employee_audit' already exists	0.000 sec
26	14:46:50	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec
27	14:46:50	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
28	14:46:50	CREATE TRIGGER after_employee_insert AFTER INSERT ON employ...	Error Code: 1359. Trigger already exists	0.016 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

```
33 (NEW.emp_id, NEW.emp_name, 'INSERT');
34 END $$
35
36 DELIMITER ;
37 • INSERT INTO employees
38 VALUES (106, 'Arjun', 1, 52000);
39 • SELECT * FROM employee_audit;
```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
*					

employees 4 employee_audit 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✗	33 14:47:21	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT...	Error Code: 1050. Table 'employee_audit' already exists	0.000 sec
✗	34 14:47:21	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec
✓	35 14:47:21	SELECT * FROM employees LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
✗	36 14:47:21	CREATE TRIGGER after_employee_insert AFTER INSERT ON employ...	Error Code: 1359. Trigger already exists	0.000 sec
✓	37 14:47:21	INSERT INTO employees VALUES (106, 'Arjun', 1, 52000)	1 row(s) affected	0.015 sec
✓	38 14:47:21	SELECT * FROM employee_audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

LocalMySQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

```
46 INSERT INTO employee_audit
47 (emp_id, emp_name, action_type)
48 VALUES
49 (OLD.emp_id, OLD.emp_name, 'DELETE');
50 END $$
51
52 DELIMITER ;
```

Result Grid

audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-15 14:47:21
NULL	NULL	NULL	NULL	NULL

employees 6 employee_audit 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
44	14:48:01	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec
45	14:48:01	SELECT * FROM employees LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
46	14:48:01	CREATE TRIGGER after_employee_insert AFTER INSERT ON employ...	Error Code: 1359. Trigger already exists	0.000 sec
47	14:48:01	INSERT INTO employees VALUES (106, 'Arjun', 1, 52000)	Error Code: 1062. Duplicate entry '106' for key 'employees.PRIMARY'	0.000 sec
48	14:48:01	SELECT * FROM employee_audit LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
49	14:48:01	CREATE TRIGGER after_employee_delete AFTER DELETE ON emplo...	0 row(s) affected	0.063 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```

48 VALUES
49 (OLD.emp_id, OLD.emp_name, 'DELETE');
50 END $$
51
52 DELIMITER ;
53 • DELETE FROM employees
54 WHERE emp_id = 106;

```

Result Grid

audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-15 14:47:21
*	NULL	NULL	NULL	NULL

employees 8 employee_audit 9 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 56	14:48:51	SELECT * FROM employees LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
✗ 57	14:48:51	CREATE TRIGGER after_employee_insert AFTER INSERT ON employ...	Error Code: 1359. Trigger already exists	0.000 sec
✗ 58	14:48:51	INSERT INTO employees VALUES (106, 'Arjun', 1, 52000)	Error Code: 1062. Duplicate entry '106' for key 'employees.PRIMARY'	0.000 sec
✓ 59	14:48:51	SELECT * FROM employee_audit LIMIT 0, 1000	1 row(s) returned	0.016 sec / 0.000 sec
✓ 60	14:48:51	CREATE TRIGGER after_employee_delete AFTER DELETE ON emplo...	Error Code: 1359. Trigger already exists	0.000 sec
✓ 61	14:48:51	DELETE FROM employees WHERE emp_id = 106	1 row(s) affected	0.015 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```

49 (OLD.emp_id, OLD.emp_name, 'DELETE');
50 END Execute the selected portion of the script or everything, if there is no selection
51
52 DELIMITER ;
53 • DELETE FROM employees
54 WHERE emp_id = 106;
55 • SELECT * FROM employee_audit;

```

Result Grid

audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-15 14:47:21
2	106	Arjun	DELETE	2026-08-15 14:48:51
3	106	Arjun	INSERT	2026-08-15 14:49:18
4	106	Arjun	DELETE	2026-08-15 14:49:18
NULL	NULL	NULL	NULL	NULL

employees 10 employee_audit 11 employee_audit 12 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
69	14:49:18	CREATE TRIGGER after_employee_insert AFTER INSERT ON employ...	Error Code: 1359. Trigger already exists	0.015 sec
70	14:49:18	INSERT INTO employees VALUES (106, 'Arjun', 1, 52000)	1 row(s) affected	0.000 sec
71	14:49:18	SELECT * FROM employee_audit LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
72	14:49:18	CREATE TRIGGER after_employee_delete AFTER DELETE ON emplo...	Error Code: 1359. Trigger already exists	0.000 sec
73	14:49:18	DELETE FROM employees WHERE emp_id = 106	1 row(s) affected	0.016 sec
74	14:49:18	SELECT * FROM employee_audit LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

LocalMySQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```

62 INSERT INTO employee_audit
63 (emp_id, emp_name, action_type)
64 VALUES
65 (NEW.emp_id, NEW.emp_name, 'UPDATE');
66 END $$
67
68 DELIMITER ;

```

Result Grid

audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-15 14:47:21
2	106	Arjun	DELETE	2026-08-15 14:48:51
3	106	Arjun	INSERT	2026-08-15 14:49:18
4	106	Arjun	DELETE	2026-08-15 14:49:18
5	106	Arjun	INSERT	2026-08-15 14:49:54
6	106	Arjun	DELETE	2026-08-15 14:49:54
7	106	Arjun	INSERT	2026-08-15 14:50:36
8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 16 employee_audit 17 employee_audit 18 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
97	14:50:36	INSERT INTO employees VALUES (106, 'Arjun', 1, 52000)	1 row(s) affected	0.016 sec
98	14:50:36	SELECT * FROM employee_audit LIMIT 0, 1000	7 row(s) returned	0.000 sec / 0.000 sec
99	14:50:36	CREATE TRIGGER after_employee_delete AFTER DELETE ON emplo...	Error Code: 1359. Trigger already exists	0.000 sec
100	14:50:36	DELETE FROM employees WHERE emp_id = 106	1 row(s) affected	0.000 sec
101	14:50:36	SELECT * FROM employee_audit LIMIT 0, 1000	8 row(s) returned	0.000 sec / 0.000 sec
102	14:50:36	CREATE TRIGGER after_employee_update AFTER UPDATE ON empl...	Error Code: 1359. Trigger already exists	0.000 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```
66 END $$
67
68 DELIMITER ;
69 • UPDATE employees
70 SET salary = 60000
71 WHERE emp_id = 101;
72 • SELECT * FROM employee_audit;
```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
	2	106	Arjun	DELETE	2026-08-15 14:48:51
	3	106	Arjun	INSERT	2026-08-15 14:49:18
	4	106	Arjun	DELETE	2026-08-15 14:49:18
	5	106	Arjun	INSERT	2026-08-15 14:49:54
	6	106	Arjun	DELETE	2026-08-15 14:49:54
	7	106	Arjun	INSERT	2026-08-15 14:50:36
	8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 22 employee_audit 23 employee_audit 24 employee_audit 25 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✖ 128	14:51:48	CREATE TRIGGER after_employee_delete AFTER DELETE ON emplo...	Error Code: 1359. Trigger already exists	0.000 sec
✔ 129	14:51:48	DELETE FROM employees WHERE emp_id = 106	1 row(s) affected	0.016 sec
✔ 130	14:51:48	SELECT * FROM employee_audit LIMIT 0, 1000	13 row(s) returned	0.000 sec / 0.000 sec
✖ 131	14:51:48	CREATE TRIGGER after_employee_update AFTER UPDATE ON empl...	Error Code: 1359. Trigger already exists	0.000 sec
✔ 132	14:51:48	UPDATE employees SET salary = 60000 WHERE emp_id = 101	0 row(s) affected Rows matched: 1 Changed: 0 Warnings: 0	0.000 sec
✔ 133	14:51:48	SELECT * FROM employee_audit LIMIT 0, 1000	14 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers* x

Limit to 1000 rows

```
70 SET salary = 60000
71 WHERE emp_id = 101;
72 • SELECT * FROM employee_audit;
73 • UPDATE employees
74 SET salary = 65000
75 WHERE emp_id = 101;
76 • SELECT * FROM employee_audit;
```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
	2	106	Arjun	DELETE	2026-08-15 14:48:51
	3	106	Arjun	INSERT	2026-08-15 14:49:18
	4	106	Arjun	DELETE	2026-08-15 14:49:18
	5	106	Arjun	INSERT	2026-08-15 14:49:54
	6	106	Arjun	DELETE	2026-08-15 14:49:54
	7	106	Arjun	INSERT	2026-08-15 14:50:36
	8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 30 employee_audit 31 employee_audit 32 employee_audit 33 employee_audit 34 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 163	14:58:17	SELECT * FROM employee_audit LIMIT 0, 1000	20 row(s) returned	0.000 sec / 0.000 sec
✗ 164	14:58:17	CREATE TRIGGER after_employee_update AFTER UPDATE ON empl...	Error Code: 1359. Trigger already exists	0.000 sec
✓ 165	14:58:17	UPDATE employees SET salary = 60000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.015 sec
✓ 166	14:58:17	SELECT * FROM employee_audit LIMIT 0, 1000	21 row(s) returned	0.000 sec / 0.000 sec
✓ 167	14:58:17	UPDATE employees SET salary = 65000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
✓ 168	14:58:17	SELECT * FROM employee_audit LIMIT 0, 1000	22 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```
87 END $$
88
89 DELIMITER ;
90 • INSERT INTO employees
91 VALUES (107, 'Ravi', 1, 15000);
92 • INSERT INTO employees
93 VALUES (108, 'Meena', 1, 30000);
```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
	2	106	Arjun	DELETE	2026-08-15 14:48:51
	3	106	Arjun	INSERT	2026-08-15 14:49:18
	4	106	Arjun	DELETE	2026-08-15 14:49:18
	5	106	Arjun	INSERT	2026-08-15 14:49:54
	6	106	Arjun	DELETE	2026-08-15 14:49:54
	7	106	Arjun	INSERT	2026-08-15 14:50:36
	8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 40 employee_audit 41 employee_audit 42 employee_audit 43 employee_audit 44 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 203	14:59:22	SELECT * FROM employee_audit LIMIT 0, 1000	29 row(s) returned	0.000 sec / 0.000 sec
✓ 204	14:59:22	UPDATE employees SET salary = 65000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
✓ 205	14:59:22	SELECT * FROM employee_audit LIMIT 0, 1000	30 row(s) returned	0.000 sec / 0.000 sec
✗ 206	14:59:22	CREATE TRIGGER check_salary BEFORE INSERT ON employees FO...	Error Code: 1359. Trigger already exists	0.000 sec
✗ 207	14:59:22	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000	0.000 sec
✓ 208	14:59:22	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	1 row(s) affected	0.016 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers x

Limit to 1000 rows

```

100 IF NEW.salary < OLD.salary THEN
101     SIGNAL SQLSTATE '45000'
102     SET MESSAGE_TEXT = 'Salary reduction is not allowed';
103 END IF;
104 END $$
105
106 DELIMITER ;

```

Result Grid

audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-15 14:47:21
2	106	Arjun	DELETE	2026-08-15 14:48:51
3	106	Arjun	INSERT	2026-08-15 14:49:18
4	106	Arjun	DELETE	2026-08-15 14:49:18
5	106	Arjun	INSERT	2026-08-15 14:49:5
6	106	Arjun	DELETE	2026-08-15 14:49:54
7	106	Arjun	INSERT	2026-08-15 14:50:36
8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 45 employee_audit 46 employee_audit 47 employee_audit 48 employee_audit 49 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 224	15:00:14	UPDATE employees SET salary = 65000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.016 sec
✓ 225	15:00:14	SELECT * FROM employee_audit LIMIT 0, 1000	35 row(s) returned	0.000 sec / 0.000 sec
✗ 226	15:00:14	CREATE TRIGGER check_salary BEFORE INSERT ON employees FO...	Error Code: 1359. Trigger already exists	0.000 sec
✗ 227	15:00:14	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000	0.000 sec
✗ 228	15:00:14	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	Error Code: 1062. Duplicate entry '108' for key 'employees.PRIMARY'	0.000 sec
✓ 229	15:00:14	CREATE TRIGGER prevent_salary_reduction BEFORE UPDATE ON e...	0 row(s) affected	0.015 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

tiggers* x

Limit to 1000 rows

```

103     END IF;
104     END $$
105
106     DELIMITER ;
107 •    UPDATE employees
108     SET salary = 40000
109     WHERE emp_id = 101;

```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
	2	106	Arjun	DELETE	2026-08-15 14:48:51
	3	106	Arjun	INSERT	2026-08-15 14:49:18
	4	106	Arjun	DELETE	2026-08-15 14:49:18
	5	106	Arjun	INSERT	2026-08-15 14:49:54
	6	106	Arjun	DELETE	2026-08-15 14:49:54
	7	106	Arjun	INSERT	2026-08-15 14:50:36
	8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 50 employee_audit 51 employee_audit 52 employee_audit 53 employee_audit 54 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 247	15:00:47	SELECT * FROM employee_audit LIMIT 0, 1000	38 row(s) returned	0.000 sec / 0.000 sec
✗ 248	15:00:47	CREATE TRIGGER check_salary BEFORE INSERT ON employees FO...	Error Code: 1359. Trigger already exists	0.000 sec
✗ 249	15:00:47	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be less than 20000	0.000 sec
✗ 250	15:00:47	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	Error Code: 1062. Duplicate entry '108' for key 'employees.PRIMARY'	0.000 sec
✗ 251	15:00:47	CREATE TRIGGER prevent_salary_reduction BEFORE UPDATE ON e...	Error Code: 1359. Trigger already exists	0.000 sec
✗ 252	15:00:47	UPDATE employees SET salary = 40000 WHERE emp_id = 101	Error Code: 1644. Salary reduction is not allowed	0.000 sec

MySQL Workbench

LocalMYSQL x

File Edit View Query Database Server Tools Scripting Help

Limit to 1000 rows

```

106 DE/ Execute the selected portion of the script or everything, if there is no selection
107 • UPDATE employees
108 SET salary = 40000
109 WHERE emp_id = 101;
110 • UPDATE employees
111 SET salary = 70000
112 WHERE emp_id = 101;

```

Result Grid

	audit_id	emp_id	emp_name	action_type	action_time
1	106	Arjun	INSERT	2026-08-15 14:47:21	
2	106	Arjun	DELETE	2026-08-15 14:48:51	
3	106	Arjun	INSERT	2026-08-15 14:49:18	
4	106	Arjun	DELETE	2026-08-15 14:49:18	
5	106	Arjun	INSERT	2026-08-15 14:49:54	
6	106	Arjun	DELETE	2026-08-15 14:49:54	
7	106	Arjun	INSERT	2026-08-15 14:50:36	
8	106	Arjun	DELETE	2026-08-15 14:50:36	

employees 55 employee_audit 56 employee_audit 57 employee_audit 58 employee_audit 59 x Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
253	15:01:15	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
254	15:01:17	CREATE DATABASE trigger_project	Error Code: 1007. Can't create database 'trigger_project'; database exists	0.000 sec
255	15:01:17	USE trigger_project	0 row(s) affected	0.000 sec
256	15:01:17	CREATE TABLE employees (emp_id INT PRIMARY KEY, emp_n...	Error Code: 1050. Table 'employees' already exists	0.000 sec
257	15:01:17	CREATE TABLE employee_audit (audit_id INT AUTO_INCREMENT...	Error Code: 1050. Table 'employee_audit' already exists	0.000 sec
258	15:01:17	INSERT INTO employees VALUES (101, 'Rahul', 1, 50000), (102, 'Priya...	Error Code: 1062. Duplicate entry '101' for key 'employees.PRIMARY'	0.000 sec



```
101      SIGNAL SQLSTATE '45000'
102      SET MESSAGE_TEXT = 'Salary reduction is not allowed';
103  END IF;
104 END $$
105
106 DELIMITER ;
107 • UPDATE employees
```

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
	2	106	Arjun	DELETE	2026-08-15 14:48:51
	3	106	Arjun	INSERT	2026-08-15 14:49:18
	4	106	Arjun	DELETE	2026-08-15 14:49:18
	5	106	Arjun	INSERT	2026-08-15 14:49:54
	6	106	Arjun	DELETE	2026-08-15 14:49:54
	7	106	Arjun	INSERT	2026-08-15 14:50:36
	8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 60 employee_audit 61 employee_audit 62 employee_audit 63 **employee_audit 64** x

Output

Action Output

	#	Time	Action	Message
✖	295	15:02:25	CREATE TRIGGER check_salary BEFORE INSERT ON employees FO...	Error Code: 1359. Trigger already e
✖	296	15:02:25	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be
✖	297	15:02:25	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	Error Code: 1062. Duplicate entry "
✖	298	15:02:25	CREATE TRIGGER prevent_salary_reduction BEFORE UPDATE ON e...	Error Code: 1359. Trigger already e
✖	299	15:02:25	UPDATE employees SET salary = 40000 WHERE emp_id = 101	Error Code: 1644. Salary reduction
✔	300	15:02:25	UPDATE employees SET salary = 70000 WHERE emp_id = 101	0 row(s) affected Rows matched: 1



109 WHERE emp_id = 101;

110 • U Execute the selected portion of the script or everything, if there is no selection

111 SET salary = 70000

112 WHERE emp_id = 101;

113 • UPDATE employees

114 SET salary = 75000

115 WHERE emp_id = 101;

	audit_id	emp_id	emp_name	action_type	action_time
▶	1	106	Arjun	INSERT	2026-08-15 14:47:21
	2	106	Arjun	DELETE	2026-08-15 14:48:51
	3	106	Arjun	INSERT	2026-08-15 14:49:18
	4	106	Arjun	DELETE	2026-08-15 14:49:18
	5	106	Arjun	INSERT	2026-08-15 14:49:54
	6	106	Arjun	DELETE	2026-08-15 14:49:54
	7	106	Arjun	INSERT	2026-08-15 14:50:36
	8	106	Arjun	DELETE	2026-08-15 14:50:36

employees 65 employee_audit 66 employee_audit 67 employee_audit 68 employee_audit 69 x

Output

Action Output

	#	Time	Action	Message
✖	320	15:03:39	INSERT INTO employees VALUES (107, 'Ravi', 1, 15000)	Error Code: 1644. Salary cannot be
✖	321	15:03:39	INSERT INTO employees VALUES (108, 'Meena', 1, 30000)	Error Code: 1062. Duplicate entry "
✖	322	15:03:39	CREATE TRIGGER prevent_salary_reduction BEFORE UPDATE ON e...	Error Code: 1359. Trigger already e
✖	323	15:03:39	UPDATE employees SET salary = 40000 WHERE emp_id = 101	Error Code: 1644. Salary reduction
✔	324	15:03:39	UPDATE employees SET salary = 70000 WHERE emp_id = 101	0 row(s) affected Rows matched: 1
✔	325	15:03:39	UPDATE employees SET salary = 75000 WHERE emp_id = 101	1 row(s) affected Rows matched: 1